Maxime GAUTIER

Born on the 01/13/1996 at Angers (France)
maxime.gautier@minesparis.psl.eu
+33 6 13 18 60 77
35 Rue de l'Amiral Mouchez, Bâtiment C, 2e étage porte au fond à droite, boîte 14, 75013, Paris

Research topics

Near surface characterization: structures and physical properties estimated with electrical resistivity tomography, seismic refraction tomography, multispectral analyses of surface waves and microgravimetry. Applied geophysics to natural hazards, active faults, landslides investigations and hydrogeophysics. Data recording, processing, inversions and interpretation.

Research experience

2024 – now
Post-thesis : Near-surface geophysics, hydrogeophysics and geostatistics. ERT, SRT and MASW geophysical imaging at the Orgeval catchment observatory site. Data acquisition, processing and analysis. Reconstruction of 3D distributions of subsurface geophysical parameters from 2D profiles using geostatistical methods. Long-term monitoring of the geophysical and hydrogeological properties of the subsurface in the context of groundwater-river interaction within an alluvial plain setting. Mines Paris – PSL / METIS Laboratory, Sorbonne University (France) (ANR GWS Bounds project led by Agnès Rivière).

2021 – 2024
3 years
PhD. In Geosciences: Electrical resistivity tomography for imaging active faults and landslides. Development of the PyMERRY code for quantifying uncertainties in ERT. Multi-data inversions of near-surface geophysics and geomorphology. At the Montpellier Geosciences Laboratory (France), Geophysical and Satellite Imaging team. Supervised by Stéphanie Gautier-Raux and Rodolphe Cattin.

2021
5 months
Final-year project: Improving hydrological models in tropical volcanic environments with new data acquisition. Application for a better understanding of flash flood phenomena in a cyclonic context across two catchment areas. At the Géosciences Réunion and OSU Réunion laboratories, Réunion Island, for the ERORUN research project. Supervised by Jean-Lambert Join and Alix Toulier.

2020
2 months
Volunteer internship: Rain-induced landslides: effects of intense rainfall on the geophysical and hydrogeochemical parameters of the Pégairolles-de-l'Escalette landslide (Hérault, France). At the Montpellier Geosciences Laboratory. Supervised by Stéphanie Gautier-Raux.

2020
3 months
R&D Project: Feasibility study of a combined drone and PIV (Particle Image Velocimetry) approach for determining the surface velocity field of a natural streams. At the Institute of Risk Sciences, IMT Mines Alès (France). Supervised by Pierre-Alain Ayral, Laurent Aprin and Pierre Lauret.

Publications (7 papers including 3 as first author)

Accepted with minor review
Gautier M., Gautier S., Cattin R. in review. Multi-method geophysical approach to assess the internal porous structure of a deep-seated slow-moving landslide. *Journal of Applied Geophysics*.

In in prep.
Toulier A., Shuler C., **Gautier M.** et al. in preparation. High-resolution gridded daily groundwater recharge estimates for Southern O'ahu, Hawai'i (1990–2024). Hydrological Processes.

In review
Thomasset C., Ritz J.F., Cathelin N., Baize S., Malcles O., **Gautier M**., Foin M. in review. Evidence for ~M6 surface-rupturing earthquake(s) during the Middle Pleistocene along the Northeastern Cévennes fault system, Marsanne segment (France). *Journal of the Geological Society*.

2026 Rivière A., **Gautier M.** et al. in press. Bridging geophysics and hydrogeology to characterize the critical zone in heterogeneous environments. *Comptes Rendus Géoscience*.

2024 **Gautier M.,** Ferry M., Gautier S. et al. 2024. Deformed alluvial terraces record an excess of slip over the last few centuries on the Himalayan Topographic Frontal Thrust of central Bhutan. Nature *Communications Earth & Environment, 5*, 590. https://doi.org/10.1038/s43247-024-01759-z

Toulier A., Join J.L., Staménoff P., Benoit Y., Lebeau G., **Gautier M.** et al. 2024. ERORUN-STAFOR: A collaborative observatory for the multidisciplinary study of the critical zone processes in a tropical volcanic watershed including a tropical montane cloud forest. Hydrological Processes, 38(2). https://doi.org/10.1002/hyp.15061

2023 **Gautier M.,** Gautier S., Cattin, R. 2023. PyMERRY: A Python solution for an improved interpretation of electrical resistivity tomography images. *Geophysics, 89*(1), F23–F39. https://doi.org/10.1190/geo2023-0105.1

Scientific communications (18 including 12 as presenter, (10 talks, 8 posters))

2026 **M. Gautier**, S. Pasquet, N. Radic, et al. *3D near-surface geophysics and geostatistics for heterogeneities characterization and water table monitoring on the Orgeval critical zone observatory*. EGU 2026 Vienna (Austria). (Poster)

N. Radic, A. Rivière, L. Bodet, A. Gesret, **M. Gautier**, et al. *Integrating Transient Hydrogeology Models for Enhanced Interpretation of Geophysical Data in the Vadose Zone*. EGU 2026 Vienna (Austria). (Talk by N. Radic)

2025 **M. Gautier**, S. Gautier, R. Cattin. *Imaging of the internal structures of a landslide using a combined gravimetry and electrical resistivity tomography approach.* RST 2025, Montpellier (France). (Oral)

M. Gautier, A. Rivière, N. Radic et al. *Geophysics and geostatistics for a 3D estimation of the physical properties of the near-surface layer of the Critical Zone.* RST 2025, Montpellier (France). (Oral)

C. Thomasset, J.F. Ritz, N. Cathelin, S. Baize, O. Macles, **M. Gautier** et M. Foin. *Quaternary activity along the Marsanne Fault revealed by paleoseismological data (SE France).* RST 2025, Montpellier (France). (Talk by C. Thomasset)

M. Gautier, S. Gautier, R. Cattin. *Imaging the internal structures of a landslide using a combined gravimetry and electrical resistivity tomography approach.* Journées aléas gravitaire 2025, Aoste (Italy). (Talk by S. Gautier)

M. Gautier, A. Rivière, N. Radic et al. *Geophysics and geostatistics for a 3D estimation of physical properties within the critical zone sub-surface.* Tereno OZCAR 3rd international conference, Paris (France). (Poster)

N. Radic, A. Rivière, V. Clinquart, A. Gesret, **M. Gautier** et al. *Long-term assessment of water and heat stream-aquifer exchanges in Avenelles Critical Zone Observatory, France (2013-2025).* Tereno OZCAR 3rd international conference, Paris (France). (Poster by N. Radic)

N. Radic, A. Rivière, A. Gesret, L. Bodet, S. Pasquet et **M. Gautier**. *Modeling the influence of the vadose zone in a geophysical study of the Orgeval critical zone observatory*. Tereno OZCAR 3rd international conference, Paris (France). (Poster by par N. Radic)

A. Rivière, L. Bodet, **M. Gautier** et al. Integrating data into the hydrogeophysical model: a case study of the Orgeval Critical Zone observatory. eLTER Science ARPHA conference 2025, Tampere (Finlande). https://doi.org/10.3897/aca.8.e152222. (Poster by A. Rivière)

M. Gautier, S. Gautier, R. Cattin. *Near-surface imaging of an active fault – An example of joint stochastic inversion of electrical, seismic, gravimetric and topographic data.* Geofcan 2025, Avignon (France). (Oral)

2023 **M. Gautier**, M. Ferry, S. Gautier, R. Cattin, R. Le Roux-Mallouf. *Unevenness of well-preserved alluvial terraces: significance and tectonic implication. Application to the Himalayan front in south-central Bhutan.* In AGU Fall Meeting 2023, San Francisco (USA). (Poster) http://dx.doi.org/10.13140/RG.2.2.23361.16482

M. Gautier, S. Gautier, R. Cattin. *PyMERRY: a Python solution for improved interpretation of electrical resistivity tomography images.* In AGU Fall Meeting 2023, San Francisco (USA). (Poster) http://dx.doi.org/10.13140/RG.2.2.10096.67845

C. Thomasset, J.-F. Ritz, N. Cathelin, K. Manchuel , R. Le Roux-Mallouf, S. Baize, S. Pouliquen, **M. Gautier**. *Cinematic and paleaseismological characteristics of the Cévennes Fault System (CFS), France*. In Earth Sciences Meeting (RST) 2023, Rennes (France). (Talk by C. Thomasset)

M. Gautier, M. Ferry, S. Gautier, R. Cattin. *Non planarity of alluvial terrasses, tectonic significations and implications.* In Earth Sciences Meeting (RST) 2023, Rennes (France). (Talk by R. Cattin)

M. Gautier, S. Gautier, R. Cattin. *A new approach to quantify the reliability of Electrical Resistivity Tomography (ERT) images.* In EGU 2023, Vienna (Austria). (Oral)

2022 **M. Gautier**, S. Gautier, R. Cattin. *A new approach of electrical sensitivity topography sensitivity for geophysical characterization of natural hazards.* In the G2 meeting 2022: Geodesy, Geophysics and Mountains, Grenoble (France). (Oral)

M. Gautier, S. Gautier, R. Cattin. *Near-surface geophysical imagery for landslides hazard.* In the Days of Landslides and Mass Movements Hazards 2022, Montpellier (France). (Poster)

Academic experiences

2025 Qualification for the post of Lecturer (CNU sections 35 and 36).

2024 – 2026 **Postdoctoral researcher,** hydrogeophysics, characterisation of the near-surface using geophysical methods (ERT, SRT seismic and MASW) for hydrogeology. Data acquisition, processing, inversion and interpretation; modelling; geostatistics; management of geophysical databases. Work carried out as part of the ANR GWS Bounds project led by Agnès Rivière. Center for Geosciences, MinesParis-PSL – METIS laboratory, Sorbonne University (France).

2021 – 2024 **PhD in Earth Sciences,** near-surface geophysical imaging for natural hazards (landslides and active faults). Estimation of uncertainties in electrical resistivity tomography images; inversions coupled with other geophysical and geomorphological data. Work carried out as part of the ANR TOPO Extrême project led by Rodolphe Cattin. *Géosciences Montpellier, Université de Montpellier (France)*

2020 – 2021 **Engineer**, Energy, Environment and Risk Department, specialising in Risk and Crisis Management Crises (GPA: 3.75/4) *École des Mines, Alès (France)*
Master's in Earth Dynamics and Natural Hazards (First Class Honours) *Université de Montpellier (France)*

2018 **Bachelor's** degree in Earth and Water Sciences (with first-class honours)*Université de Montpellier (France)*

2014 – 2016 Preparatory classes for French Grandes Écoles, Biology Chemistry Physics Earth Sciences *Lycée Le Fresne, Angers (France)*

Teaching, supervision and scientific activities (94 in total)

The dissemination of knowledge is an integral part of a scientist's remit and has therefore been a feature of my career from the very beginning: during my PhD, I undertook two additional teaching assignments of 32 hours each (University of Montpellier), followed by part-time teaching posts during my postdoctoral research (Mines Paris - PSL, ENS-Paris). The teaching hours and types of courses are broken down as shown in the table below. These courses covered the following levels and topics:

	TD	TP	Terrain	
Université de Montpellier (2022-2024)	6h L2(3h), L3(3h)	30h L1(13h), L2(12h), M1(5h)	30h L1(3h), L2(18h), L3(9h)	66h
ENS – Paris (2025-2026)	18h M2(18h)	/		10h
Mines Paris – PSL (2025)	10h 1A(10h)	/		94h
	34h	30h	30h	94h

- BSc in Earth Sciences 1st year: general geology (igneous, sedimentary and metamorphic rocks, dating, fossils, isostasy, fieldwork).
- BSc in Earth Sciences 2nd year: geological mapping, fieldwork, GIS (QGIS).
- BSc in Earth Sciences 3rd year: hydrology, flow measurement in natural watercourses, data acquisition and processing.
- MSc Natural Hazards : ERT geophysical inversions and gravimetry (Python, PyGIMLi).
- 1A Civil Engineering Cycle (Mines Paris - PSL): (seismic refraction, data acquisition and processing).
- MSc in Geosciences 2ne year, Water Resources Modelling (ENS Paris): hydrogeological data inversion using Bayesian methods.

I have also been involved in supervising and assessing the following internship projects:

- Member of the assessment panel for the M2 Natural Hazards module of the Geosciences Master's programme and examiner for three students (University of Montpellier, 2022) .
- Co-supervision of two M2 internships (6 months) on the hydrogeophysics of the Orgeval catchment (Mines Paris - PSL, UMR METIS Sorbonne University, 2025 and 2026).
- Co-supervision of a six-month Master's (M1) internship in gravimetry at the Pégairolles-de-l'Escalette site (University of Montpellier, 2023).

Scientific coordination and responsibilities:

- Organisation and coordination of the scientific field trip for the TERENO-OZCAR conference (Paris 2025) at the Avenelles observatory site. Presentation of the geophysical equipment and methods used, as well as the preparation of interactive 3D visual aids to share the results obtained in terms of multi-method characterisation of the near-surface (ERT, SRT, MASW).
- Elected PhD student representative (2021–2022): participation in the executive committee of the Géosciences Montpellier laboratory.